

CIS 115 Quick Review and Quick Start

Chapters 3 and 4

After naming the item in an IPO chart, a(n) _____ must be chosen.

C Data Type	
char	■
short	■ ■
int	■ ■ ■ ■
float	■ ■ ■ ■
double	■ ■ ■ ■ ■ ■ ■ ■

After naming the item in an IPO chart, a(n) Datatype must be chosen.

C Data Type	
char	■
short	■ ■
int	■ ■ ■ ■
float	■ ■ ■ ■
double	■ ■ ■ ■ ■ ■ ■ ■

In C++, the two types of memory locations available for you to temporarily store data are _____ and _____.



In C++, the two types of memory locations available for you to temporarily store data are variables and named constants.



The value of $a(n)$ _____ is permitted to change during the execution of a program.

$n + 3$



a number plus three

the variable

The value of a(n) variable is permitted to change during the execution of a program.

n + 3



a number plus three

the variable

A(n) _____ is also known as a reserved word.



A(n) keyword is also known as a reserved word.



The last rule in naming identifiers/variables in C++ is that all identifiers/variables are case _____.



The last rule in naming identifiers/variables in C++ is that all identifiers/variables are case sensitive.



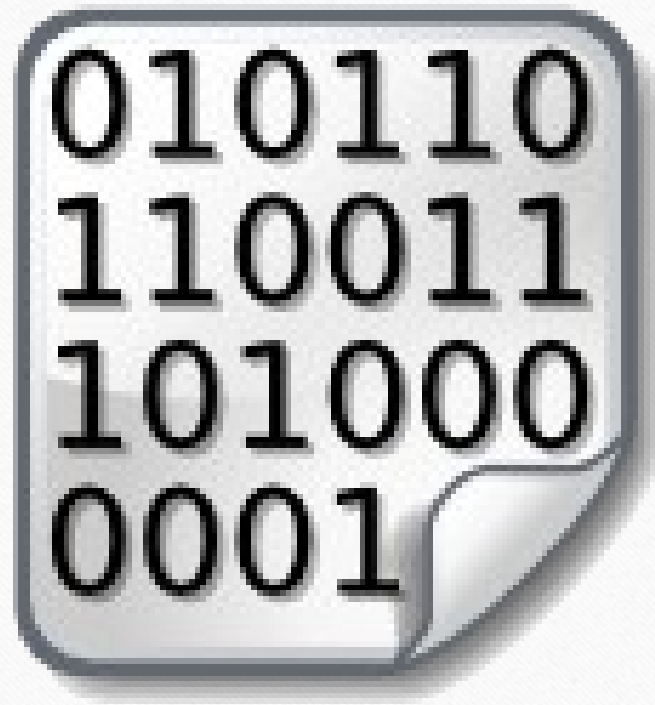
The string data type is not one of the _____ data types in C++.



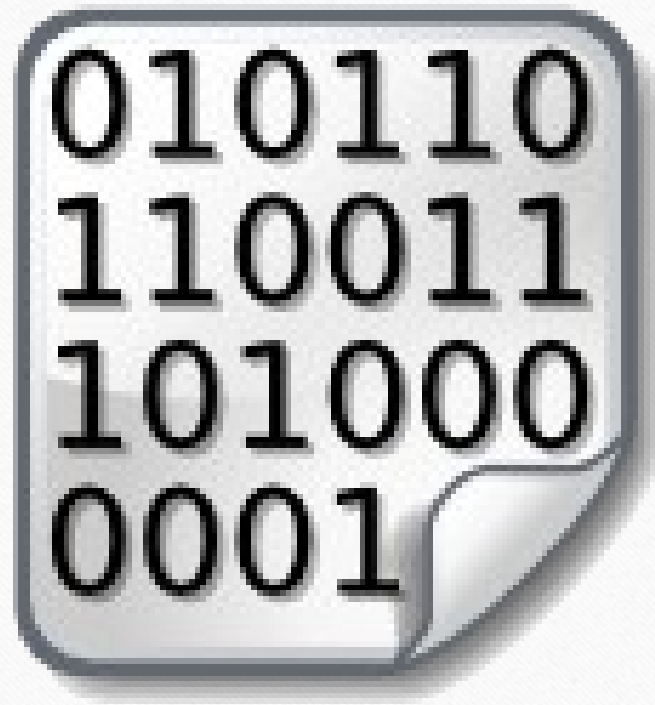
The string data type is not one of the fundamental data types in C++.



Numbers are represented in internal memory using the _____ number system.

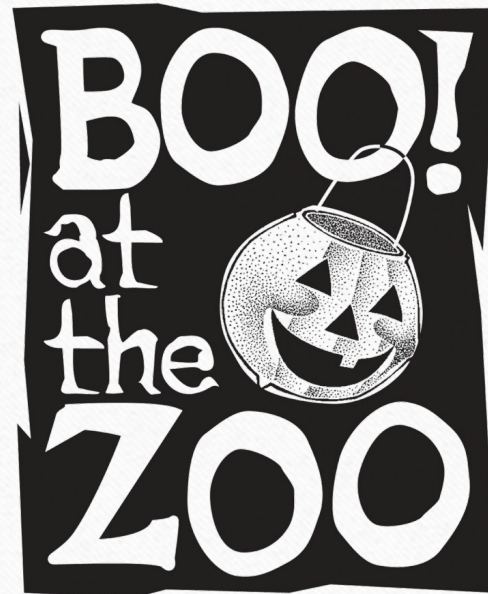


Numbers are represented in internal memory using the binary number system.



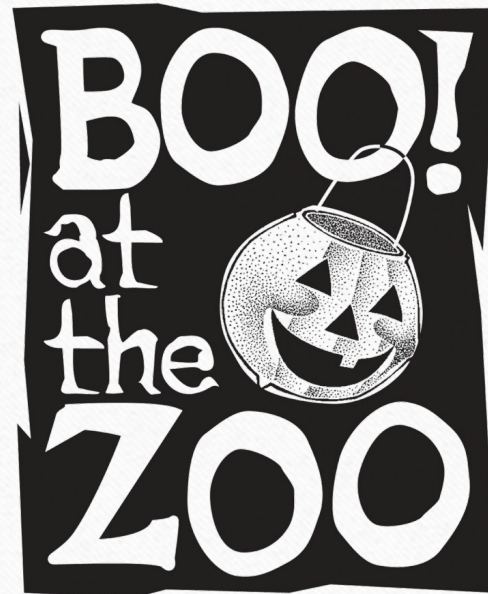
Variables of type _____ can take on the values of true and false.

 U.S. Cellular presents



Variables of type Boolean datatype can take on the values of true and false.

 U.S. Cellular presents



The float and double data types store _____ numbers.



The float and double data types store real numbers.



Unlike variables and named constants, _____ constants are not memory locations.

LITERAL



Unlike variables and named constants, literal constants are not memory locations.



A **string** literal constant is zero or more characters enclosed in _____ quotation marks.



A **string** literal constant is zero or more characters enclosed in _double quotation marks.



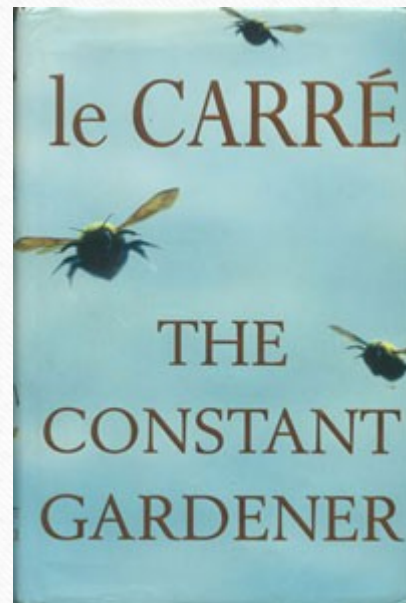
A **character** literal constant is a single character enclosed in _____
quotation marks.



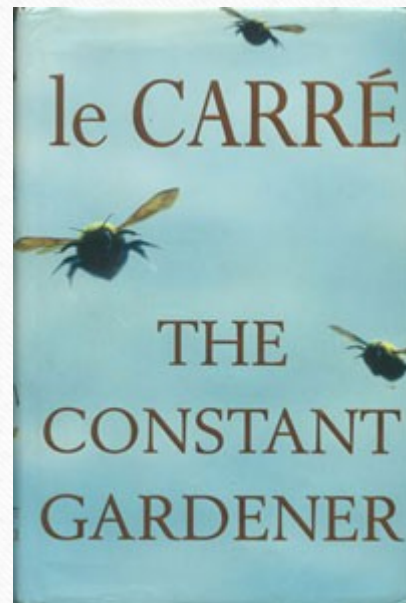
A **character** literal constant is a single character enclosed in single quotation marks.



The _____ keyword is used to indicate that a memory location is a named constant.



The constant keyword is used to indicate that a memory location is a named constant.



All statements in C++ must end with a(n) _____.

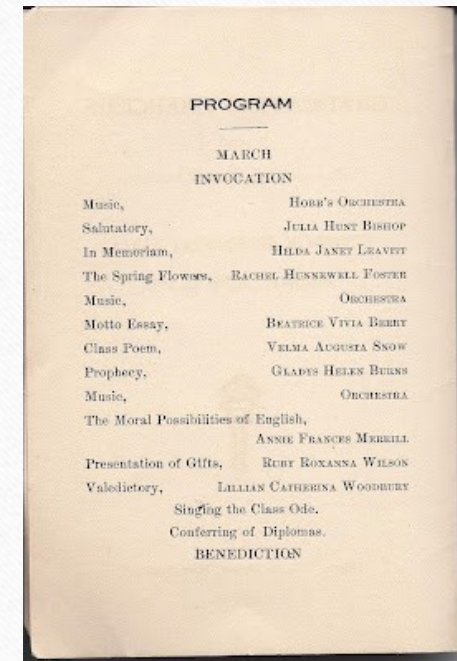


All statements in C++ must end with a(n) _semicolon.

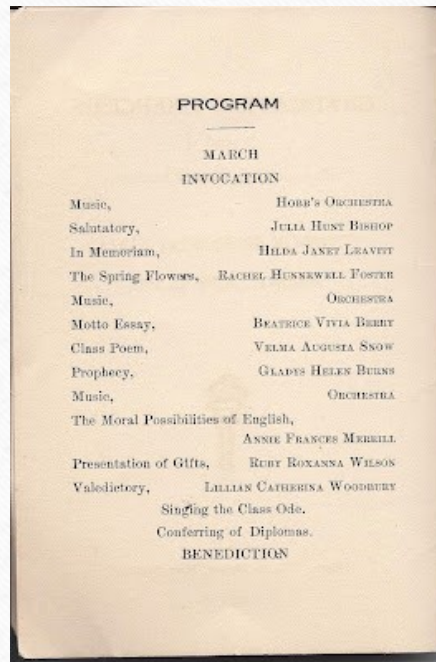


Chapter 4

- Still in the 4th step of the problem-solving process.
- What is the 4th step?



Answer: Coding



A photograph of a printed program page, likely from a school or organizational event. The page is titled "PROGRAM" and lists various activities and the names of the participants or performers. The text is centered and formatted in a simple, serif font. The page is slightly aged and has a light beige background.

PROGRAM	
MARCH	
INVOCATION	
Music,	Hobb's ORCHESTRA
Salutatory,	JULIA HUNT BISHOP
In Memoriam,	HEIDA JANET LEAVITT
The Spring Flowers,	RACHEL HUNNEWELL FOSTER
Music,	ORCHESTRA
Motto Essay,	BEATRICE VIVIAN BERRY
Class Poem,	VELMA AUGUSTA SNOW
Prophecy,	GLADES HELEN BURNS
Music,	ORCHESTRA
The Moral Possibilities of English,	ANNIE FRANCES MERRILL
Presentation of Gifts,	RUBY ROXANNA WILSON
Valedictory,	LILLIAN CATHERINE WOODBURY
Singing the Class Ode.	
Conferring of Diplomas.	
BENEDICTION	

Input and Output Statements

- What they are and how they work.



Input and Output Statements Answer

- Input statements bring data from the screen into the program.
- Output statements send data from the program to the screen.

Objects Perform ...

- Input and Output Operations
- Called Stream Objects
- A **stream** is a sequence of characters



What is an **Object**?

Answer: What is an **Object**?

- Performs standard input and output operations

Why are they called **stream** objects?

Answer: Why are they called **stream objects**?

They handle streams of characters.



What is a **stream**?

Answer: What is a **stream**?

- A sequence of characters.



More About Objects

- Created by a **Class**
- Resides in a special area of the computer's internal memory
- A **class** is simply a group of instructions used to create an object

How are Objects Created?

Answer: How are Objects Created?

- By a class



What is a **class**?

Answer: What is a **class**?

- A group of instructions used to create an object.



cin

- Gets information **into** the program
- Used with an **extraction** operator
- >>
- Allows the user to communicate with the program while it is running

When does the extraction stop?

- When the cin object encounters a **white-space character**.
- A white-space character:
 - Newline (Enter)
 - Tab
 - Spacebar key (blank)

What is the extraction operator?



Answer: What is the extraction operator?

• >>

How is a white-space character created?

Answer: How is a white-space character created?

- Newline (Enter)
- Tab
- Spacebar (blank character)

Examples

- `cin >> item;`
- `cin >> salary;`
- `cin >> number;`
- `cin >> letter;`

What happens after the **cin** statement?

- The computer waits for the user to enter something on the screen.
- The cursor just sits and blinks until something is entered.
- The program is on “pause”.

But...

- We can't just say `cin >>`
- Because the user will only see a blinking cursor

We need a **Prompt**

- A **message** that tells the user what to enter.

cout

- Used with an **insertion** operator
- <<
- Displays information on the screen
- To “send out” information to the user

When do you use **cin**?

Answer: When do you use **cin**?

- To get information into the program from a user.

When do you use **cout**?

Answer: When do you use **cout**?

- To send information from the program to the screen.



Stream Manipulator

- Allows you to **manage** the characters in the input or output stream
- We will use **endl** with cout
- **endl** moves the cursor to the next line

cout Examples

- `cout << salary;`
- `cout << number;`
- `cout << amount << endl;`
- `cout << "This is your salary " << salary;`
- `cout << "This is your number " << number;`
- `cout << "This is the amount " << amount << endl;`

What does the **insertion operator** look like?



INSERT COIN

What does the **insertion operator** look like? Answer:

- <<

What does the **extraction operator** look like?



What does the **extraction operator** look like? Answer

- >>

//

-
- Used for comments
 - All good programmers use comments
 - //declare variables
 - //calculate the total
 - //display the output
 - //program name, purpose, date

IDE

- Integrated Development Environment
- Visual Studio 2019

